Ranking Flaky Test Failures with Build Histories and Change Distance

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Flaky tests pass and fail on the same code. This paper ranks the failures of a build by the chance that each failure is flaky. On 4,210 failures of ten projects, the ranking places the flaky failure first in 71% of the builds.

CCS Concepts: • **Software and its engineering → Software testing and debugging**.

Additional Key Words and Phrases: flaky tests, regression testing, continuous integration

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1 Introduction

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[1]The build logs come from https://api.github.com/repos/apache/commons-lang/actions/runs?per_page=100&page=1.

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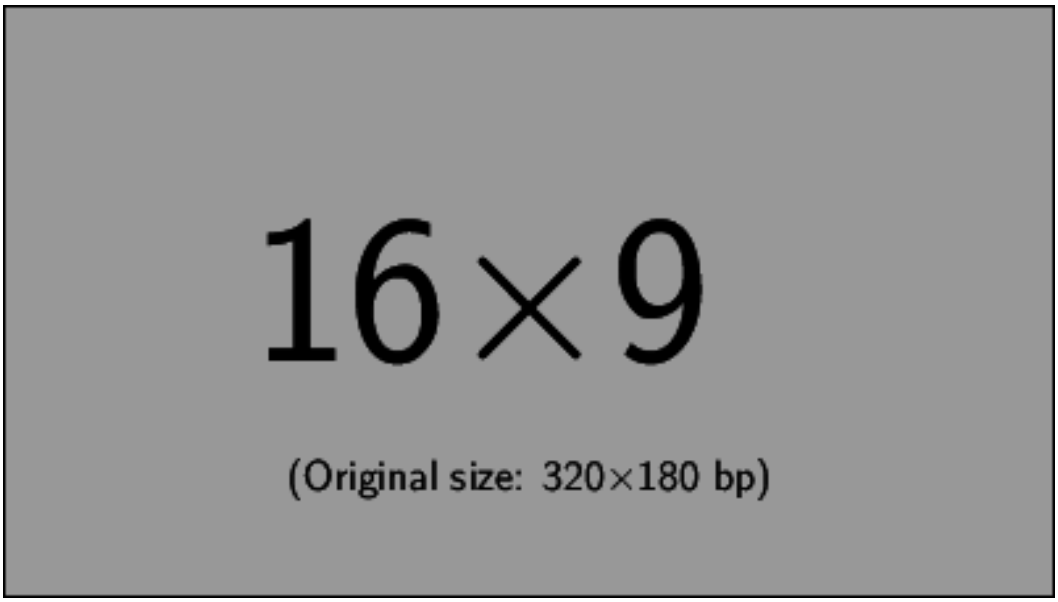


Fig. 1. A build page as a raster image of the full text width.

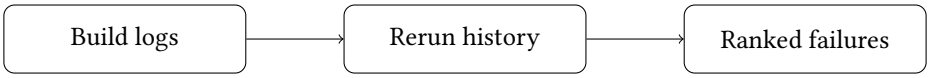


Fig. 2. The stages of the ranking.

2 Approach

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Listing 1. A test that depends on the order of the suite.

```
@Test
public void readsSharedCache() {
    assertEquals(0, SharedCache.getInstance().size());
    SharedCache.getInstance().put("key", computeExpensiveValueFromTheNetwork("https://example.org/a
        /long/path"));
}
```

[2]A second footnote with a long address, https://docs.github.com/en/actions/using-workflows/events-that-trigger-workflows#workflow_run.

Table 1. Rank of the flaky failure per project and method

Method	lang	guava	jsoup	spring	netty	kafka	flink	hadoop	hbase	Total
Ranking	61	133	30	140	88	120	95	160	110	937
Rerun order	33	70	16	77	41	64	52	85	62	500

3 Evaluation

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Data Availability

The data and scripts are available at https://anonymous.4open.science/r/flaky-ranking-replication-package-2027/README.md.

References

[1] Wing Lam, Reed Oei, August Shi, Darko Marinov, and Tao Xie. 2019. iDFlakies: A Framework for Detecting and Partially Classifying Flaky Tests. In *Proc. ICST*. 312–322.

[2] GitHub. 2026. Events that trigger workflows. https://docs.github.com/en/actions/writing-workflows/choosing-when-your-workflow-runs/events-that-trigger-workflows